

16HMIC2

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June 9, 2026

Problem

Let ABC be an acute triangle with circumcenter O , orthocenter H , and circumcircle Ω . Let M be the midpoint of AH and N the midpoint of BH . Assume the points M, N, O, H are distinct and lie on a circle ω . Prove that the circles ω and Ω are internally tangent to each other.

Solution. Radius of ω is half of radius of Ω due to homothety. Since the radius of Ω is equal to the circumradius of $\triangle ABC$, we know that the diameter of ω is equal to the circumradius of $\triangle ABC$. Since O lies on ω , this concludes that ω and Ω are internally tangent.